

A comment on "A practical significance bias in laypeople's evaluation of scientific findings"

Jan Philipp Röer, Thomas Ostermann, and Sebastian Appelbaum

Witten/Herdecke University, Department of Psychology and Psychotherapy, Witten,
Germany

Corresponding Author: Jan Philipp Röer, jan.roeer@uni-wh.de

Abstract

We tested the computational reproducibility of a research claim from Michal and Shah (2024) according to which individuals assume that a scientific finding described in a media report is meaningfully large when numerical information about the magnitude of this finding is missing. This reflects the following statement from the paper's abstract: "Participants who viewed interventions with unreported effect magnitudes were more likely to endorse interventions compared with those who viewed interventions with small effects and were just as likely to endorse interventions as those who viewed interventions with large effects, suggesting a practical significance bias.". Using the dataset from the original study, we successfully reproduced the focal test of the paper for both studies. We uncovered a couple of errors in the process, which did not change the studies' main results. This reproduction was conducted during the replication games workshop at the META-REP Conference 2024 in Munich, Germany.

1 Introduction

The to-be-reproduced paper (Michal & Shah, 2024) deals with the question how people estimate the size of effects they read about in the media and what influence this has on everyday decisions. Specifically, the authors investigated whether numerical information about the magnitude of a finding helps people to arrive at better decisions. Two studies were conducted to test this. In both studies, participants were randomly assigned to one of four different conditions (No effect size information, Large effect, Small effect no prompt, Small effect with prompt). In Study 1, participants read an article describing a new math curriculum to boost elementary students' math proficiency. In Study 2, participants read an article about a new energy bar with beneficial effects on mental alertness.

1.1 Outcome Measures

In both studies, participants rated the meaningfulness of the improvement from the intervention in relation to its cost on a scale from 0 (not at all) to 100 (very much). In Study 1, participants were asked to what extent they would recommend the new math curriculum on a scale from 0 (do not recommend) to 100 (strongly recommend). In Study 2, participants rated how likely they were to purchase the new energy bar (Study 2) on a scale from 0 (very unlikely) to 100 (very likely). Participants also completed the Abbreviated Numeracy Scale (Weller et al., 2013).

1.2 Focal Test

After controlling for numeracy, decision ratings should be higher for participants in the no effect size information condition compared to the small effect no prompt condition and decision ratings should not differ between the no effect size information condition and the large effect condition.

1.3 Reproducibility criteria

The statistical analysis was the same as in the original study. We ran an analysis of variance (ANOVA) using a regression model with decision rating as the dependent variable, numeracy as a continuous covariate, and condition as a between-subjects factor to reproduce the focal test.

2 Computational Reproducibility Results

For the computational reproduction, we used the dataset from the original study retrieved from the OSF project page at <https://osf.io/4927f/>. The raw data were provided, but no cleaning or analysis code.

2.1 Study 1

2.1.1 Descriptive statistics

The reported and calculated mean values were the same in the subgroups “No effect size information” and “Large effect”, while there were slight differences in the “Small effect no prompt” and “Small effect with prompt” groups (Table 1.1). We highlighted the discrepancies between the calculated values and the values reported in the paper in yellow.

Table 1.1. Descriptive statistics for Study 1 of Michal & Shah (2024).

Condition	Sample size	Reported Mean Values	Calculated Mean Values
No effect size information	$N = 97$	$M = 80.56$	$M = 80.56$
Large effect	$N = 101$	$M = 73.93$	$M = 73.93$
Small effect no prompt	$N = 100$	$M = 45.26$	$M = 45.18$
Small effect with prompt	$N = 100$	$M = 45.22$	$M = 45.02$

2.1.2 F-values

The reported and calculated *F*-values in Table 1.2 were identical for both parameters, while there was a small difference in the η^2 of numeracy.

Table 1.2. *F*-values for Study 1 of Michal & Shah (2024).

Parameter		df	<i>F</i> -value	<i>p</i> -value	η^2
Numeracy	<i>Reported</i>	1,393	16.58	< .001	.04
	<i>Calculated</i>	1,393	16.58	< .001	.02
Condition	<i>Reported</i>	3,393	49.46	< .001	.27
	<i>Calculated</i>	3,393	49.46	< .001	.27

2.1.3 Mean differences and effect sizes

Table 1.3 summarizes our reproduction results with respect to the mean differences and to the effect sizes. We found differences to the reported values, due to the fact that the authors did not report the method how to calculate the mean differences. In our computational reproduction, we used Tukey's Honest Significant Difference (HSD) and marginal means [Mar. Means] to calculate and compare the mean differences.

Table 1.3. Mean differences and effect sizes for Study 1 of Michal & Shah (2024).

Comparsion		Mean diffe- rence	95% CI	p-value	Cohen' s d	BF
No effect size information vs. small effect no prompt	<i>Reported</i>	34.83	[25.18, 44.48]	< .001	1.41	7.09 × 10 ¹⁵
	<i>HSD</i>	35.38	[25.58, 45.18]	< .001	1.41	7.09 × 10 ¹⁵
	<i>Mar. Means</i>	34.89	[25.23, 44.55]	< .001		
No effect size information vs. large effect	<i>Reported</i>	6.00	[-3.63, 15.63]	.38	0.32	1.62
	<i>HSD</i>	6.63	[-3.15, 16.40]	.30	0.32	1.62
	<i>Mar. Means</i>	6.07	[-3.57, 15.70]	.37		
Small effect no prompt vs. large effect	<i>Reported</i>	28.83	[19.27, 38.38]	< .001	1.06	3.23 × 10 ⁹
	<i>HSD</i>	28.75	[19.05, 38.45]	< .001	1.06	3.23 × 10 ⁹
	<i>Mar. Means</i>	28.82	[19.26, 38.37]	< .001		
Small effect with prompt vs. small effect no prompt	<i>Reported</i>	0.02	[-9.56, 9.59]	1.00	0.005	0.15
	<i>HSD</i>	0.16	[-9.57, 9.89]	1.00	0.005	0.15
	<i>Mar. Means</i>	0.004	[-9.58, 9.58]	1.00		

2.1.4 Interaction analysis

We used the model Decision Rating ~ Numeracy + Condition + Numeracy × Condition to test for a statistical interaction between numeracy and condition and found the same results as in the original study (Table 1.4).

Table 1.4. Interaction analysis for Study 1 of Michal & Shah (2024).

Parameter		df	F-value	p-value	η ²
Numeracy	<i>Reported</i>	3,390	3.69	= .01	.03
	<i>Calculated</i>	3,390	3.69	= .01	.03

2.2 Study 2

2.2.1 Descriptive statistics

In Study 2, the reported and calculated mean values were the same (Table 2.1).

Table 2.1. Descriptive data for Study 2 of Michal & Shah (2024).

Condition	Sample size	Reported Mean Values	Calculated Mean Values
no effect size information	$N = 100$	$M = 50.47$	$M = 50.47$
large effect	$N = 99$	$M = 58.90$	$M = 58.90$
small effect no prompt	$N = 100$	$M = 34.98$	$M = 34.98$
small effect with prompt	$N = 101$	$M = 31.68$	$M = 31.68$

2.2.2 F-values

The reported and calculated F-values in Table 1.2 were identical for both parameters, while there was a minor difference in the η^2 of condition.

Table 2.2. F-values for Study 1 of Michal & Shah (2024).

Parameter		df	F-value	p-value	η^2
Numeracy	<i>Reported</i>	1,395	17.78	< .001	.04
	<i>Calculated</i>	1,395	17.78	< .001	.04
Condition	<i>Reported</i>	3,395	19.18	< .001	.13
	<i>Calculated</i>	3,395	19.18	< .001	.12

2.2.3 Mean differences and effect sizes

With respect to the mean differences and effect-sizes, Table 2.3 summarizes our reproduction results. Again, we found differences to the reported values, both due to the fact that the authors did not report the method how to calculate the mean differences and also due to missing signs. We used Tukey's Honest Significant Difference (HSD) and marginal means [Mar. Means] to calculate and compare the mean differences in the computational reproduction.

Table 2.3. Mean differences and effect sizes for Study 2 of Michal & Shah (2024)

Comparsion		Mean difference	95% CI	p-value	Cohen's d	BF
No effect size information vs. small effect no prompt	<i>Reported</i>	14.75	[4.13, 25.36]	.002	0.54	115.47
	<i>HSD</i>	15.49	[4.67, 26.31]	.001	0.54	115.47
	<i>Mar. Means</i>	14.76	[4.14, 25.39]	.002		
No effect size information vs. large effect	<i>Reported</i>	7.52	[3.12, 18.17]	.26	0.29	1.08
	<i>HSD</i>	8.43	[-2.42, 19.28]	.19	0.29	1.08
	<i>Mar. Means</i>	7.45	[-3.11, 18.20]	.26		
Small effect no prompt vs. large effect	<i>Reported</i>	22.27	[11.63, 32.91]	< .001	0.83	497754.3
	<i>HSD</i>	23.92	[13.07, 34.77]	< .001	0.83	497754.3
	<i>Mar. Means</i>	22.31	[11.62, 33.00]	< .001		
Small effect with prompt vs. small effect no prompt	<i>Reported</i>	5.20	[5.39, 15.79]	0.59	0.11	0.20
	<i>HSD</i>	3.30	[-7.50, 14.09]	0.59	0.11	0.20
	<i>Mar. Means</i>	5.15	[-5.50, 15.81]	0.60		

2.2.4 Interaction analysis

Again, we used the model Decision Rating ~ Numeracy + Condition + Numeracy × Condition to test for a statistical interaction between numeracy and condition and found the same results as in the original study (Table 2.4).

Table 2.4. Interaction analysis for Study 2 of Michal & Shah (2024).

Parameter		df	F-value	p-value	η^2
Numeracy	<i>Reported</i>	3,392	3.13	= .026	.02
	<i>Calculated</i>	3,392	3.13	= .026	.02

2.2.5 Correlations

The reported and calculated correlations were the same with a small difference in the “Small effect with prompt” condition.

Table 2.5. Correlations for Study 2 of Michal & Shah (2024).

Condition	Reported correlations (r,95% CI)	Calculated correlations (r,p)
No effect size information	<.1	<.1
Large effect	<.1	<.1
Small effect no prompt	-.19 [-.37, -.006]	-.19 [-.37, -.007]
Small effect with prompt	-.41 [-.57, -.24]	-.42 [-.57, -.24]

2.3 Discrepancies Between Pre-analysis Plan and Article

The authors registered the study and published pre-analysis plans for each study which are available on the OSF project page at <https://osf.io/4927f/>. While we found no deviation from the pre-analysis plan and the article in a strict sense, a second pre-analysis plan was

published for Study 1 that deviates from the first pre-analysis plan. We highlighted the discrepancies between the first and the second pre-analysis plan in yellow.

First pre-analysis plan

- (1) Recommendation ratings will not be statistically different between the Large Effect and No Effect Size conditions (non-directional)
- (2) Recommendation ratings for the Small Effect No Prompt condition will be significantly lower than both the Large Effect and No Effect Size conditions (directional)
- (3) Recommendation ratings for the Small Effect With Prompt condition will be significantly lower than the Small Effect No Prompt condition (directional)

Second pre-analysis plan

- (1) Recommendation ratings will not be significantly different for participants in the No Effect Size and Large Effect Size conditions, after controlling for numeracy ability (non-directional)
- (2) Recommendation ratings will be significantly lower for participants in the Small Effect No Prompt than the No Effect Size condition, after controlling for numeracy ability (directional)
- (3) Recommendation ratings will be significantly lower for participants in the Small Effect With Prompt than the Small Effect No Prompt condition, after controlling for numeracy ability (directional).

According to the authors, the first pre-analysis plan was published prior to the creation of data (March 24 2023) and the second pre-analysis plan was published prior to human observation of the data (April 3 2023). No explanation is given for the changes.

3 Conclusion

During the replication games workshop at the META-REP Conference 2024, we have dedicated an entire day to computationally reproduce the focal tests for Study 1 and Study 2 from Michal and Shah (2024). We are happy to report that the reproduction was largely successful in that we only found minor to moderate errors that do not challenge the authors' core conclusions. Altogether, the scope and implications of these errors do not seem to necessitate a correction of the article.

We believe that some deviations could have been avoided if the authors—unless we have overlooked something—had reported how exactly the mean differences were calculated (2.1.3 and 2.2.3). For the sake of transparency and openness, it would be ideal if this information were available in order to determine whether the results can be reproduced independently using the same method. Other minor errors we have found are missing signs leading to incorrect confidence intervals (2.2.3) and incorrect values in the text ("Study 1 no prompt $M = 45.26$; Study 2 no prompt $M = 34.98$; Study 2 with prompt $M = 45.22$; Study 2 with prompt $M = 31.68$ ", p. 320).

Furthermore, when reviewing the OSF registrations, we noticed that two different pre-analysis plans were uploaded for Study 1, which deviate in three out of four hypotheses (2.3). A major advantage of preregistration is that it can be made transparent which hypotheses and analyses were determined by the researchers prior to data collection and which were determined by the data (Lakens, 2024; Willroth & Atherton, 2024; see also Nosek, Ebersole, DeHaven & Mellor, 2018). The authors state that the second pre-analysis plan was published prior to any human observation of the data. We have no

reason to doubt that the information provided by the authors is correct, but ideally all deviations from the pre-analysis plan should be transparently reported, especially if the changes are somewhat extensive and are published after the data collection has already been completed.

References

- Michal, A. L., & Shah, P. (2024). A practical significance bias in laypeople's evaluation of scientific findings. *Psychological Science*, 35(4), 315–327. <https://doi.org/10.1177/09567976241231506>
- Lakens, D. (2024). When and how to deviate from a preregistration. *Collabra: Psychology*, 10(1), 117094. <https://doi.org/10.1525/collabra.117094>
- Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. (2018). The preregistration revolution. *Proceedings of the National Academy of Sciences of the United States of America*, 115(11), 2600–2606. <https://doi.org/10.1073/pnas.1708274114>
- Weller, J. A., Dieckmann, N. F., Tusler, M., Mertz, C. K., Burns, W. J., & Peters, E. (2013). Development and testing of an abbreviated numeracy scale: A Rasch analysis approach. *Journal of Behavioral Decision Making*, 26(2), 198–212. <https://doi.org/10.1002/bdm.1751>
- Willroth, E. C., & Atherton, O. E. (2024). Best laid plans: A guide to reporting preregistration deviations. *Advances in Methods and Practices in Psychological Science*, 7(1). <https://doi.org/10.1177/25152459231213802>